AN IDEAL SUPPORTED ON REGULAR 2-DIMENSIONAL POINTS BECOMES INVERTIBLE AFTER POINT BLOWUPS

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Editorial note. The scheme-case principalization hypothesis (regular 2-dimensional local ring at the support) holds on each U_i by construction.

Allowed standard prerequisites (editorial).

- Principalization of supported ideals on regular surfaces (schemes). Resolution of Surfaces, lemma make-ideal-principal (cross-chapter, named). Produces the blowup sequence on each etale neighbourhood making I_i invertible.
- Blowup sequences under the etale neighbourhood equivalence. lemma-equivalence-sequence-blowups (carded L26828). Assembles the per-neighbourhood sequences into one global sequence of blowups of X.

Source reference. Original lemma 0BHF: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 507–507. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BHG: focused-seeds/geometry/S9-spres-make-ideal-principal/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 0BH9: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 391–391. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Let $T \subset |X|$ be a finite set of closed points x such that (1) X is regular at x and (2) the local ring of X at x has dimension 2. Let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent sheaf of ideals such that $\mathcal{O}_X/\mathcal{I}$ is supported on T. Then there exists a sequence*

$$X_m \to X_{m-1} \to \ldots \to X_1 \to X_0 = X$$

where $X_{j+1} \to X_j$ is the blowing up of X_j at a closed point x_j lying above a point of T such that $\mathcal{I}\mathcal{O}_{X_n}$ is an invertible ideal sheaf.

Date: stacks-project:spaces-resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Say $T = \{x_1, \ldots, x_r\}$. Pick elementary étale neighbourhoods $(U_i, u_i) \to (X, x_i)$ as in Section 0BH9. For each i the restriction $\mathcal{I}_i = \mathcal{I}|_{U_i} \subset \mathcal{O}_{U_i}$ is a quasi-coherent sheaf of ideals supported at u_i. The local ring of U_i at u_i is regular and has dimension 2. Thus we may apply Resolution of Surfaces, Lemma 0BHG to find a sequence

$$X_{i,m_i} \to X_{i,m_i-1} \to \ldots \to X_1 \to X_{i,0} = U_i$$

of blowing ups in closed points lying over u_i such that $\mathcal{I}_i\mathcal{O}_{X_{i,m_i}}$ is invertible. By Lemma 0BHF we find a sequence of blowing ups

$$X_m \to X_{m-1} \to \ldots \to X_1 \to X_0 = X$$

as in the statement of the lemma whose base change to our U_i produces the given sequences. It follows that $\mathcal{I}\mathcal{O}_{X_n}$ is an invertible ideal sheaf. Namely, we know this is true over $X \setminus \{x_1, \ldots, x_n\}$ and in an étale neighbourhood of the fibre of each x_i it is true by construction. $\square$